

Clase 2: Orientación a Objetos

Fer Cargnelutti

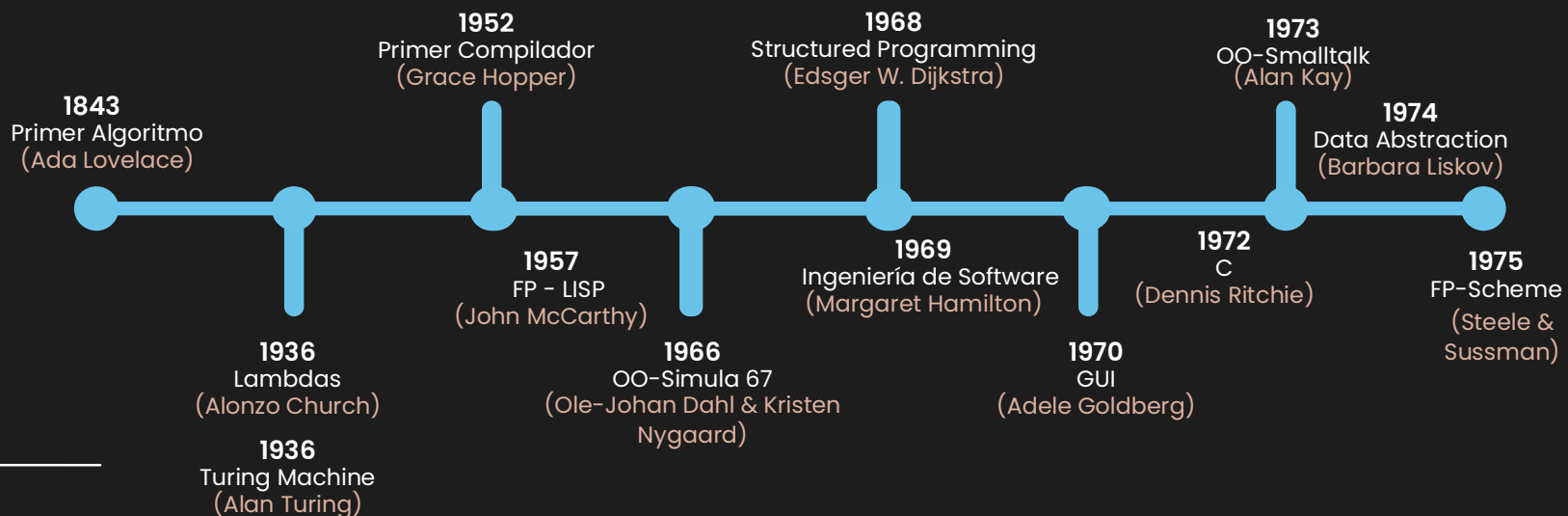
R E C A P

clean code

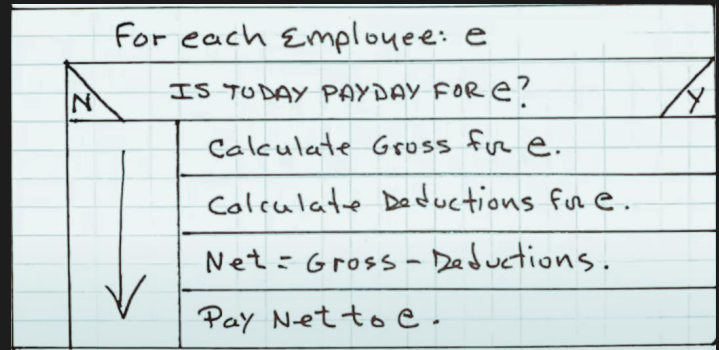
- El código es el **principal** documento de diseño
- El **mal** código no se comenta, se arregla!
- Usar nombres que se puedan decir y buscar.
- Métodos: 1 sola cosa, 3-4 líneas.
- La duplicación es el origen de todos los males



Un poco de historia



Structured paradigm



Principles

- ✓ Sequences
- ✓ Selections
- ✓ Iterations

GOTO



Structured paradigm

```
public void processPayroll(List<Employee> employees) {
    LocalDate today = LocalDate.now();

    for (Employee e : employees) {
        if (isPayday(today, e)) {
            double gross = calculateGross(e);
            double deductions = calculateDeductions(e);
            double net = gross - deductions;

            payNetToEmployee(e, net);
        }
    }
}
```

Principles

- ✓ Sequences
- ✓ Selections
- ✓ Iterations

GOTO



Functional paradigm



```
double calculateProductValue(Producto producto) {  
    return producto.getPrecio() * producto.getStock();  
}
```

```
double calculateFilteredInventoryValue(List<Producto> products, Predicate<Producto> filter) {  
    final var filteredProducts = filterProducts(products, filter);  
    final var filteredProductValues =  
        transformProducts(filteredProducts, this::calculateProductValue);  
    final double totalFilteredInventoryValue = sumValues(filteredProductValues);  
    return totalFilteredInventoryValue;  
}
```

Principles

- ✓ Pure functions
- ✓ Immutable variables
- ✓ First-Class and Higher-Order Functions
- ✓ Recursion
- ✓ Referential transparency

Object Oriented Paradigm

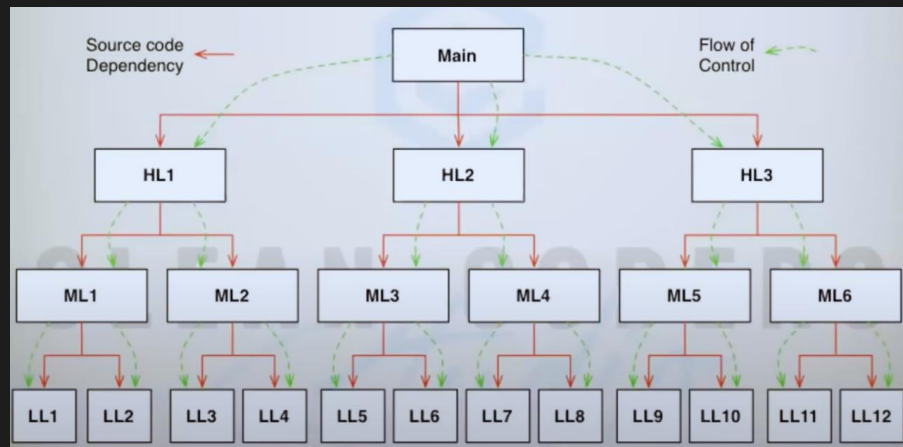


```
account.debit(100);
```

Principles

- ✓ Abstraction
- ✓ Encapsulation
- ✓ Inheritance
- ✓ Polymorphism

Dependencies



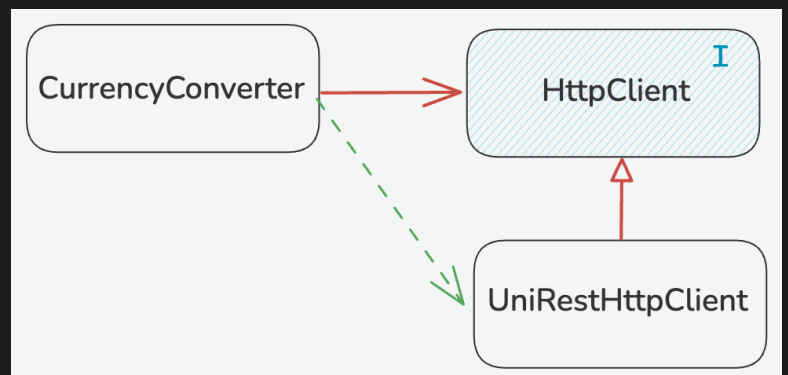
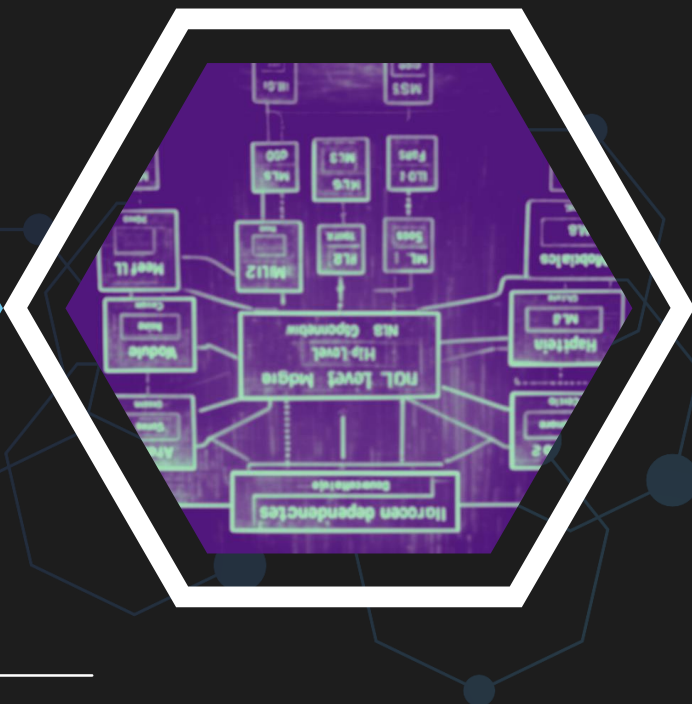
SOURCE CODE
DEPENDENCIES



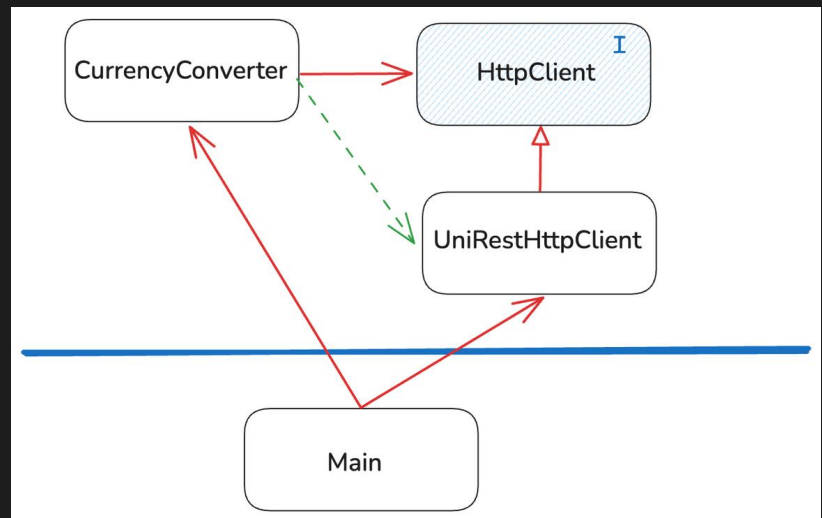
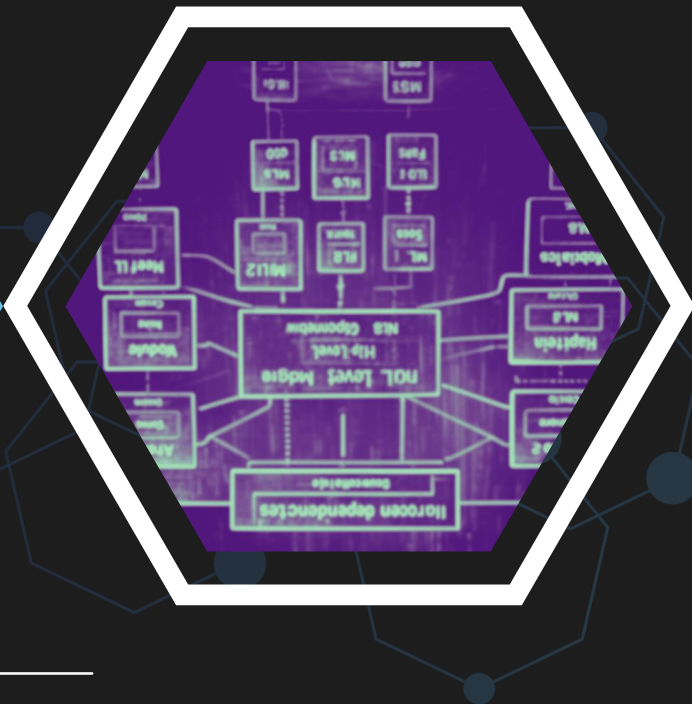
FLOW OF
CONTROL



Dependencies



Dependencies



FLOW OF
CONTROL



SOURCE CODE
DEPENDENCIES



What NOT to do...



Functional

Discipline imposed upon
assignment (mutability)

Object Oriented

Discipline imposed upon
indirect transfer of control

Structured

Discipline imposed upon
direct transfer of control

